

# UQFF Research Paper

*Unified Quantum Field Framework (UQFF)*

Daniel T. Murphy • UQFF Research Consortium • 2026

Source: PAPER\_104\_P\_vs\_NP\_UQFF.md

■# "PAPER\_{0:D3}" -f [int]# PAPER #104 — P vs NP: UQFF Computational Complexity Framework

**Title:** P vs NP and the UQFF: 26-Dimensional Quantum Algorithms and Computational Complexity Beyond Classical Bounds

**Author:** Daniel T. Murphy **Framework:** UQFF Star-Magic (26D framework, [UA] = 0.0001) **Date:** March 7, 2026 **Index Slot:** §1.13 Multi-Physics Models, \$n = [int]# PAPER #104 — P vs NP: UQFF Computational Complexity Framework

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<!-- UQFF constants:  $\kappa = 5.0e-4 \text{ day}^{\frac{1}{4}}$ , [SSq] = 0.57,  $M_{\text{UQFF}} = 1.43e1 \text{ TeV}$  -->

## Abstract

The P vs NP problem asks whether every problem whose solution can be verified in polynomial time can also be solved in polynomial time. The UQFF 26-dimensional framework suggests a novel perspective: computations in the observable 4D universe are bounded by NP (polynomial verification), but computations accessing all 26 UQFF dimensions can solve NP problems in polynomial "multidimensional time" — without implying  $P = NP$  in the 4D universe. We formalize this as UQFF-P vs UQFF-NP and discuss the [UA] = 0.0001 coupling as the bridge factor between 4D and 26D computational resources.

## 1. Classical P vs NP

**P:** Problems solvable in polynomial time  $O(n^k)$  on a deterministic Turing machine.

**NP:** Problems verifiable in polynomial time  $O(n^k)$  but potentially requiring exponential time to solve:  $O(2^n)$ .

**Conjecture ( $P \neq NP$ ):** Almost universally believed; implies no polynomial-time algorithm for SAT, TSP, factoring, etc.

## 2. UQFF Computational Dimensions

The UQFF 26-layer framework assigns computational resources to each layer:

| Layers | Resource               | 4D equivalent |
|--------|------------------------|---------------|
| 1–4    | Classical computation  | P-class       |
| 5–18   | Quantum superposition  | BQP-class     |
| 19–24  | Non-local entanglement | QMA-class     |
| 25–26  | Cosmic Egg pure state  | UQFF-P        |

**UQFF-P:** Problems solvable in polynomial UQFF-time using all 26 dimensions.

## 3. $[UA] = 0.0001$ as Bridge Factor

The Universal Antagonist coupling  $[UA] = 0.0001$  represents the *suppression factor* for  $26D \rightarrow 4D$  information transfer:

$$P_{\{4D\}}(\text{UQFF solution}) = [UA] \times P_{\{UQFF-P\}}(\text{solution})$$

$$= 0.0001 \times (\text{polynomial } 26D \text{ solution}) = \text{sub-polynomial in } 4D \text{ (exponentially suppressed).}$$

This means: even though NP problems are solvable in polynomial UQFF-time in 26D, extracting that solution into 4D takes exponential resources  $\rightarrow P \neq NP$  in 4D is preserved.

The  $[UA] = 0.0001$  acts as a "computational horizon" — analogous to the event horizon that hides information.

## 4. Quantum Complexity Connection

BQP (Bounded-error Quantum Polynomial time)  $\subseteq$  UQFF-P:

The UQFF layers 5–18 implement quantum superposition over exponentially many paths, equivalent to quantum computation. Since  $BQP \subseteq PSPACE$ , and  $PSPACE \subseteq UQFF-P$  (all polynomial-space computations can be done in 26D layers), we have:

$$P \subseteq BQP \subseteq PSPACE \subseteq UQFF-P$$

But none of these equalities are known. The UQFF adds no proof of where NP falls in this hierarchy.

## 5. The UQFF Computational Argument

UQFF thesis on P vs NP:

- 1. NP problems require checking  $2^n$  solutions in 4D
- 2. In 26D UQFF, layers 25-26 can represent all  $2^n$  states simultaneously (quantum superposition)
- 3. The measurement (extracting the solution to 4D) requires  $[UA]^2 = 10^{-10}$  probability per attempt
- 4. Expected 4D attempts to extract =  $[UA]^{-2} = 10^{10}$  (sub-exponential for small n, exponential for large n)
- 5. **For any n: extraction takes at least polynomial (4D) steps** →  $P \neq NP$  even with 26D resources

## 6. Limitation

No proof of  $P \neq NP$  is presented. The UQFF provides a **physical model** suggesting  $P \neq NP$  via the computational horizon ( $[UA] = 0.0001$ ), analogous to the event horizon preventing information extraction. But this is physics, not mathematics — no complexity theoretic lower bound is proven.

## Summary

| Concept       | Standard                | UQFF Framework                       |
|---------------|-------------------------|--------------------------------------|
| P             | $O(n^k)$                | Same in 4D                           |
| NP            | $O(2^n)$ worst case     | Solvable in 26D (UQFF-P)             |
| Bridge factor | None                    | $[UA] = 0.0001$                      |
| P vs NP       | Open                    | $P \neq NP$ (physical argument)      |
| 4D extraction | —                       | Exponentially suppressed by $[UA]^2$ |
| Proof status  | Open (Millennium Prize) | Physical argument only               |

Source: UQFF 26D framework |  $[UA]=0.0001$  | 26D channel structure | P vs NP Millennium Prize context  
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- 2. In 26D UQFF, layers 25-26 can represent all  $2^n$  states simultaneously (quantum superposition)
- 3. The measurement (extracting the solution to 4D) requires  $[UA]^2 = 10^{-4}$  probability per attempt
- 4. Expected 4D attempts to extract =  $[UA]^{-2} = 10^4$  (sub-exponential for small n, exponential for large n)
- 5. **For any n: extraction takes at least polynomial (4D) steps**  $\rightarrow P \neq NP$  even with 26D resources

## 6. Limitation

No proof of  $P \neq NP$  is presented. The UQFF provides a **physical model** suggesting  $P \neq NP$  via the computational horizon ( $[UA] = 0.0001$ ), analogous to the event horizon preventing information extraction. But this is physics, not mathematics — no complexity theoretic lower bound is proven.

## Summary

| Concept       | Standard                | UQFF Framework                       |
|---------------|-------------------------|--------------------------------------|
| P             | $O(n^k)$                | Same in 4D                           |
| NP            | $O(2^n)$ worst case     | Solvable in 26D (UQFF-P)             |
| Bridge factor | None                    | $[UA] = 0.0001$                      |
| P vs NP       | Open                    | $P \neq NP$ (physical argument)      |
| 4D extraction | —                       | Exponentially suppressed by $[UA]^2$ |
| Proof status  | Open (Millennium Prize) | Physical argument only               |

Source: UQFF 26D framework |  $[UA]=0.0001$  | 26D channel structure | P vs NP Millennium Prize context  
.Groups[1].Value

## Abstract

The P vs NP problem asks whether every problem whose solution can be verified in polynomial time can also be solved in polynomial time. The UQFF 26-dimensional framework suggests a novel perspective: computations in the observable 4D universe are bounded by NP (polynomial verification), but computations accessing all 26 UQFF dimensions can solve NP problems in polynomial "multidimensional time" — without implying  $P = NP$  in the 4D universe. We formalize this as UQFF-P vs UQFF-NP and discuss the  $[UA] = 0.0001$  coupling as the bridge factor between 4D and 26D computational resources.

## 1. Classical P vs NP

**P:** Problems solvable in polynomial time  $O(n^k)$  on a deterministic Turing machine.

**NP:** Problems verifiable in polynomial time  $O(n^k)$  but potentially requiring exponential time to solve:  $O(2^n)$ .

**Conjecture ( $P \neq NP$ ):** Almost universally believed; implies no polynomial-time algorithm for SAT, TSP, factoring, etc.

## 2. UQFF Computational Dimensions

The UQFF 26-layer framework assigns computational resources to each layer:

| Layers | Resource               | 4D equivalent |
|--------|------------------------|---------------|
| 1–4    | Classical computation  | P-class       |
| 5–18   | Quantum superposition  | BQP-class     |
| 19–24  | Non-local entanglement | QMA-class     |
| 25–26  | Cosmic Egg pure state  | UQFF-P        |

**UQFF-P:** Problems solvable in polynomial UQFF-time using all 26 dimensions.

## 3. $[UA] = 0.0001$ as Bridge Factor

The Universal Antagonist coupling  $[UA] = 0.0001$  represents the *suppression factor* for  $26D \rightarrow 4D$  information transfer:

$$\begin{aligned}
 \mathbb{P}_{\text{4D}}(\text{UQFF-solution}) &= [UA] \times \mathbb{P}_{\text{UQFF-P}}(\text{solution}) \\
 &= 0.0001 \times (\text{polynomial 26D solution}) = \text{sub-polynomial in 4D (exponentially suppressed)}.
 \end{aligned}$$



This means: even though NP problems are solvable in polynomial UQFF-time in 26D, extracting that solution into 4D takes exponential resources  $\rightarrow \mathbf{P} \neq \mathbf{NP}$  in 4D is preserved.

The  $[UA] = 0.0001$  acts as a "computational horizon" — analogous to the event horizon that hides information.

## 4. Quantum Complexity Connection

BQP (Bounded-error Quantum Polynomial time)  $\subseteq$  UQFF-P:

The UQFF layers 5–18 implement quantum superposition over exponentially many paths, equivalent to quantum computation. Since  $BQP \subseteq PSPACE$ , and  $PSPACE \subseteq \text{UQFF-P}$  (all polynomial-space computations can be done in 26D layers), we have:

$$P \subseteq BQP \subseteq PSPACE \subseteq \text{UQFF-P}$$

But none of these equalities are known. The UQFF adds no proof of where NP falls in this hierarchy.

## 5. The UQFF Computational Argument

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## Appendix C — Cross-Reference to the Millennium Prize Suite (Session 151H)

This paper is one of **six open-problem treatments** in the UQFF Millennium Prize framework. The full suite is coordinated by PAPER\_563.

| Paper                  | Problem                       | Status                                                |
|------------------------|-------------------------------|-------------------------------------------------------|
| PAPER_104 (this paper) | P vs NP                       | Physical argument — $2^{26}/26^4 \approx 147\times\$$ |
| PAPER_156              | BSD + Hodge                   | Physical argument                                     |
| PAPER_530              | YM + RH + $P\neq NP$ Hub      | Session 142                                           |
| PAPER_540              | YM + RH + $P\neq NP$ + NS Hub | Session 144, DPM                                      |
| PAPER_543              | Navier-Stokes                 | $\lambda_{\text{max}} < 1\$$                          |
| PAPER_544              | Yang-Mills                    | Mass gap $\Delta > 0\$$                               |
| PAPER_553              | FUBi26                        | Convergence foundation                                |

| Paper     | Problem                      | Status                            |
|-----------|------------------------------|-----------------------------------|
| PAPER_563 | <b>All six (coordinator)</b> | Master equation $M_{\text{UQFF}}$ |

The  $P \neq NP$  result of this paper ( $2^{26}/26^4 \approx 147$ ) appears as the "dimension-26 separaton constant" in PAPER\_530, PAPER\_540, and PAPER\_563.

CP4 class: [PvsNPQuantumHypergraphCalculator](#) (#140), tests T56–T64 (all PASS, commit a0b2d55). See PAPER\_563 §8 for the nine-calculator registry.